Synthetic Smart Wiper System - High-Level Design

SYNTHETIC TEST DATA - fictional example for tool testing. Not an official AUTOSAR document and not Tata Technologies material.

Document ID: SYN-HLD-WIPER-001. Version: 1.1.

1 Introduction

1.1 Purpose

This synthetic High-Level Design describes a fictional Smart Wiper system. It exists only to test the AutoDoc AI tool.

1.2 Scope

The scope covers rain detection, wiper speed selection, motor driving and fault storage.

2 System Overview

2.1 Software Components

WiperControlSwc coordinates the wiper speed request.

RainSensorSwc measures rain intensity and provides it through RainSensorInterface.

WiperMotorDriverSwc drives the wiper motor and reports faults.

VehicleSpeedProviderSwc supplies the vehicle speed.

DiagnosticHandlerSwc stores wiper faults.

WiperPositionSensorSwc measures the wiper blade position for WiperControlSwc.

3 Interfaces and Ports

3.1 RainSensorInterface

The port P_RainIntensity is provided by RainSensorSwc.

The port R_RainIntensity is required by WiperControlSwc.

The signal Sig_RainIntensity is updated every 20 ms.

3.2 MotorCommandInterface

The port P_MotorCommand is provided by WiperControlSwc.

The port R_MotorCommand is required by WiperMotorDriverSwc.

The signal Sig_MotorSpeedCmd carries the requested motor speed.

3.3 SpeedInterface

The signal Sig_VehicleSpeed is provided by VehicleSpeedProviderSwc.

RainSensorIf is the legacy name used by the diagnostic tooling.

4 Functional Flows

4.1 Automatic Wiping Flow

First, RainSensorSwc sends Sig_RainIntensity to WiperControlSwc every 50 ms.

Then WiperControlSwc requests the current vehicle speed from VehicleSpeedProviderSwc.

Next, WiperControlSwc sends Sig_MotorSpeedCmd to WiperMotorDriverSwc.

4.2 Fault Handling Flow

If WiperMotorDriverSwc detects an overcurrent, it reports a fault to DiagnosticHandlerSwc.

After a fault is reported, DiagnosticHandlerSwc notifies WasherPumpSwc and stores the fault.

5 Requirements

HLD-FUN-001: WiperControlSwc shall select the wiper speed from the rain intensity and the vehicle speed.

HLD-FUN-002: WiperMotorDriverSwc shall limit the motor current to 10 A.

HLD-FUN-002: DiagnosticHandlerSwc shall store each wiper fault.

HLD-FUN-004: WiperPositionSensorSwc shall report the blade position.